

Name: _____

Date: _____

PROBABILITY, (TREE DIAGRAMS)

LO: I can draw and use tree diagrams to calculate probabilities of combined events.

Tree diagrams

A tree diagram shows all possible outcomes of two or more events. Each branch represents a probability, and branches from the same point sum to 1.

Rules: Multiply along branches (AND). Add between paths (OR).

Quick examples

●	Multiply along branches for AND	$P(A \text{ and } B) = P(A) \times P(B)$
●	Add separate paths for OR	different routes
●	All branches from a point sum to 1	exhaustive
●	Add along branches for AND	must multiply
●	Multiply between paths for OR	must add

Key idea

Multiply along branches (AND), add between paths (OR).
Each set of branches sums to 1.

$$P(A \text{ and } B) = P(A) \times P(B | A)$$

Along = multiply, Between = add

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - two coins

Two fair coins are flipped. Find $P(\text{two heads})$.

$$P(H) = \frac{1}{2} \text{ each time}$$

$$P(HH) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$= \frac{1}{4}$$

Multiply along the HH branch.

Example 2 - at least one

Two coins: find $P(\text{at least one head})$.

Paths with at least one H: HH, HT, TH

$$P = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

$$= \frac{3}{4}$$

Or use $1 - P(\text{no heads}) = 1 - \frac{1}{4} = \frac{3}{4}$.

Example 3 - without replacement

Bag: 3 red, 2 blue. Pick 2 without replacement. $P(\text{both red})$?

$$P(\text{1st red}) = \frac{3}{5}$$

$$P(\text{2nd red} \mid \text{1st red}) = \frac{2}{4} = \frac{1}{2}$$

$$P(RR) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$$

$$= \frac{3}{10}$$

Without replacement changes the second probability.

Example 4 - independent events

$P(\text{rain Monday}) = 0.3$, $P(\text{rain Tuesday}) = 0.4$. $P(\text{rain both days})$?

Independent events: multiply
 $P(\text{rain Mon AND rain Tue}) = 0.3 \times 0.4 = 0.12$

$$= 0.12$$

Independent means one does not affect the other.

YOUR TURN – PRACTICE (deliberate practice)

1. Two-coin problems

Two fair coins are tossed. Find each probability.

- a) P(both tails)
- b) P(one head, one tail)
- c) P(at least one tail)
- d) P(no tails)

2. Die and coin

A coin is tossed and a die is rolled. Find:

- a) P(head and 6)
- b) P(tail and even)
- c) P(head and prime)
- d) P(tail and less than 3)

3. Without replacement

Bag: 4 red, 6 blue. Two drawn without replacement.

- a) P(both red)
- b) P(both blue)
- c) P(one of each colour)
- d) P(at least one red)

4. Real-world tree diagrams

P(bus late) = 0.2, P(train late) = 0.1. Independent.

- a) P(both late)
- b) P(neither late)
- c) P(exactly one late)
- d) P(at least one late)

5. Mixed tree diagram problems

Draw a tree diagram for each and calculate.

a) 3 coins: P(all heads)

b) 3 coins: P(exactly 2 heads)

c) Bag: 5 red, 3 blue. Two picks WITH replacement. P(same colour) both

d) P(win) = 0.6. Two games. P(win

e) P(win) = 0.6. Two games. P(win exactly one)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Multiply along branches for AND ☐

Correct answer: _____

Reason: _____

b) Add along branches for combined events ☐

Correct answer: _____

Reason: _____

c) Without replacement changes the denominator ☐

Correct answer: _____

Reason: _____

d) P(at least one) = 1 - P(none) ☐

Correct answer: _____

Reason: _____

e) With and without replacement give the same answers ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A bag has **5 red** and **4 blue** counters. Two are drawn without replacement. Find P(same colour) and P(different colours). Verify they sum to 1.

Expression: _____

Simplified: _____

b) A football team has P(win) = **0.5**, P(draw) = **0.3**, P(lose) = **0.2**. They play **2 games**. Find P(6 points), P(4 points), and P(0 points). [Win = 3pts, Draw = 1pt, Lose = 0pts]

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

